

Flappy Bird AI - How to Run

Step-by-step guide for <https://github.com/Anish-185/FlappyBirdAI>

This project uses a **genetic algorithm** to evolve a small neural network that plays Flappy Bird. The repository already contains every training log (606 runs), so you can watch the trained bird and rebuild all results within a minute. Retraining from scratch is optional and takes about 45 minutes.

You only need to type the commands in the grey boxes. Lines starting with # are comments.

Step 1 - Install the prerequisites

You need **Python 3.10 or newer** and **git**. Check them in a terminal:

```
python --version      # or: python3 --version
git --version
```

If either is missing, install it:

System	Command
Ubuntu / Debian	<code>sudo apt install python3 python3-venv git fonts-liberation</code>
Arch / Omarchy	<code>sudo pacman -S python git ttf-liberation</code>
Fedora	<code>sudo dnf install python3 git liberation-serif-fonts liberation-sans-fonts liberation-mono-fonts</code>
macOS	<code>brew install python git</code>
Windows	Install Python from python.org (tick <i>Add python.exe to PATH</i>) and Git from git-scm.com

The Liberation fonts are only needed for Step 8 (building the paper PDF).

Step 2 - Download the project

```
git clone https://github.com/Anish-185/FlappyBirdAI.git
cd FlappyBirdAI
```

Every command after this one is run inside the FlappyBirdAI folder.

Step 3 - Create a virtual environment

A virtual environment keeps this project's libraries separate from the rest of your system. Many Linux systems block `pip install` outside one.

```
python -m venv .venv
```

Then **activate** it. Do this again every time you open a new terminal:

```
source .venv/bin/activate      # Linux / macOS
.venv\Scripts\activate        # Windows
```

Your prompt now starts with `(.venv)`.

Step 4 - Install the libraries

```
pip install -r requirements.txt
```

This installs numpy, scipy, pandas, matplotlib, numba (makes the game simulator fast), reportlab (writes PDFs), pillow and pygame-ce (the game window). It takes a minute or two.

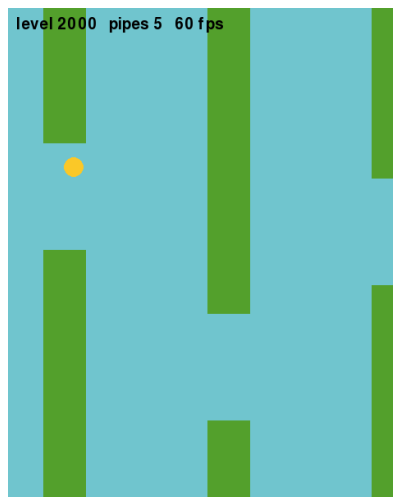
Check that everything installed:

```
python -c "import numpy, scipy, pandas, matplotlib, numba, reportlab, pygame; print('ok')"
```

Step 5 - Watch the trained bird play

```
python play.py
```

A window opens and the best evolved bird flies through a level it never saw during training. The top line shows the level number and the pipes cleared so far.



Key	Action
SPACE or mouse click	Next level (there are 30 unseen test levels)
UP / DOWN	Double / halve the speed
ESC	Quit

You can watch the champion of any experiment, and set the starting speed:

```
python play.py rule_ga          # the 2-gene hand-designed rule
python play.py rs               # random search - much worse
python play.py pool_1          # trained on only 1 level
python play.py base --fps=240  # start 4x faster
```

Step 6 - Reproduce the statistics

```
python analysis.py
```

This takes about 5 seconds. It prints one table per study and saves `results.pkl`. The most important columns:

Column	Meaning
median, q1, q3	Pipes cleared on the 30 unseen test levels (median and quartiles over 12 runs)

Column	Meaning
surv	Fraction of test levels survived for the full 10,000 frames
delta	Cliff's delta effect size vs. the reference condition (-1 to +1)
p, p_holm	Mann-Whitney U p-value, raw and after Holm correction (below 0.05 = significant)
auc, e50	Area under the learning curve; evaluations needed to reach 50 pipes

Step 7 - Regenerate the figures

```
python figures.py
```

This takes about 7 seconds and rewrites the 11 images in `figures/`. Open them with any image viewer.

Step 8 - Rebuild the paper

```
python build_paper.py
```

This takes about 3 seconds and writes `Flappy_Bird_Neuroevolution_Paper.pdf`. The warning *uncited refs: ['deb', 'boxcar']* is harmless.

This step needs the Liberation fonts from Step 1. On macOS or Windows, edit the `FD` line in `paperlib.py` to point at a folder containing `LiberationSerif-Regular.ttf`.

Step 9 - (Optional) Retrain everything from scratch

`experiments.py` skips any experiment that already has a log, so move the shipped logs out of the way first:

```
mv logs logs_shipped          # Windows: ren logs logs_shipped
python experiments.py          # about 45 minutes on one core
```

Progress is written to `logs/progress.txt`. If you stop it, run the same command again and it continues where it left off. Other options:

```
python experiments.py base rs  # train only these experiments (~1 min each)
python experiments.py --secs=600 # stop after about 10 minutes
```

Then repeat Steps 6-8 to see results from your own runs. The paper's text has its numbers typed in, so after retraining the figures change but the text does not.

Expect 56 of the 60 experiments to match the shipped logs exactly. The 4 fitness-sharing experiments (`div_share1-4`) depend on your CPU's math library and may come out different. See the *Reproducibility note* in `README.md`.

Troubleshooting

Problem	Fix
<code>error: externally-managed-environment</code>	You skipped Step 3. Create and activate the virtual environment, then retry.
<code>ModuleNotFoundError: No module named 'numpy' (or similar)</code>	The virtual environment is not active. Run the activate command from Step 3.
<code>ModuleNotFoundError: No module named 'game'</code>	Run commands from inside the <code>FlappyBirdAI</code> folder (Step 2).

Problem	Fix
TTFError: Can't open file ... LiberationSerif-Regular.ttf	Install the Liberation fonts (Step 1) or edit FD in paperlib.py (Step 8).
experiments.py finishes instantly	All logs already exist. See Step 9.
play.py: no window appears	It needs a desktop. On a remote server use Steps 6-8 instead.
First run of a script is slow	numba is compiling the simulator. Later runs use the cache.

Command summary

```
git clone https://github.com/Anish-185/FlappyBirdAI.git && cd FlappyBirdAI
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python play.py           # watch the bird
python analysis.py       # statistics
python figures.py        # figures/
python build_paper.py    # paper PDF
python experiments.py    # retrain (only after moving logs/ aside)
```